

FreeToken on 4 GiB — The Picture Book

Qwen3.6-35B-A3B-NVFP4 on an RTX 3050 Laptop, in diagrams

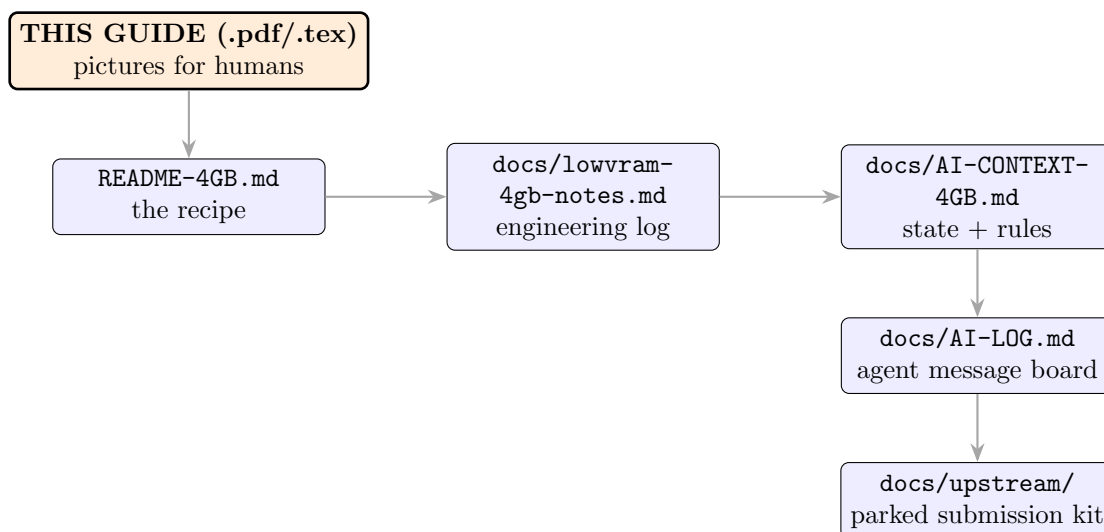
FreeToken4GB fork — visual companion to docs/

2026-09-14

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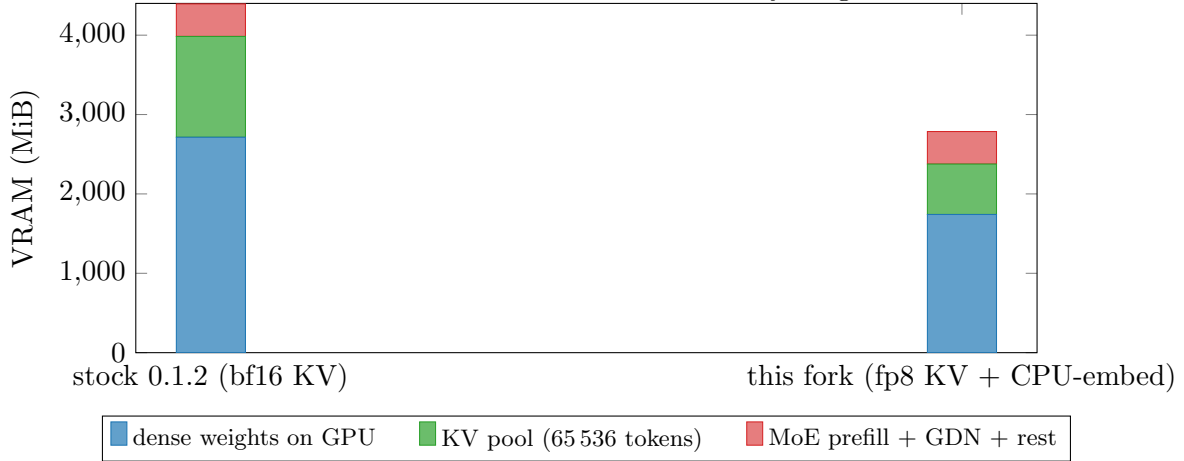
1 You are here: the reading map



Each layer adds detail: the recipe says *what to run*; the notes say *why it works and what was measured*; the AI files say *where the project stands and what is next*.

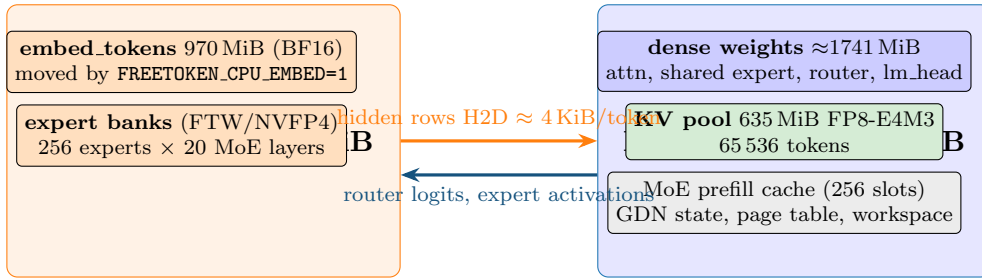
2 The budget problem: why 4 GiB was “impossible”

Stock FreeToken 0.1.2 needs more than the card has. With everything in BF16:



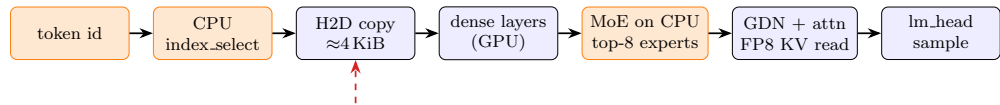
Stock: $2714 + 1270 + 410 \approx 4394 \text{ MiB} > 4096$ — over the card before CUDA-graph scratch.
 This fork: embedding table (970 MiB) moved to CPU and KV halved to FP8 fits with **717.8 MiB to spare** after full init.

3 Where the model lives: the two-machine-in-one view



Decode MoE runs on 8 CPU threads pinned to cores 0–3; the GPU keeps attention, the dense path and sampling. One card, two brains.

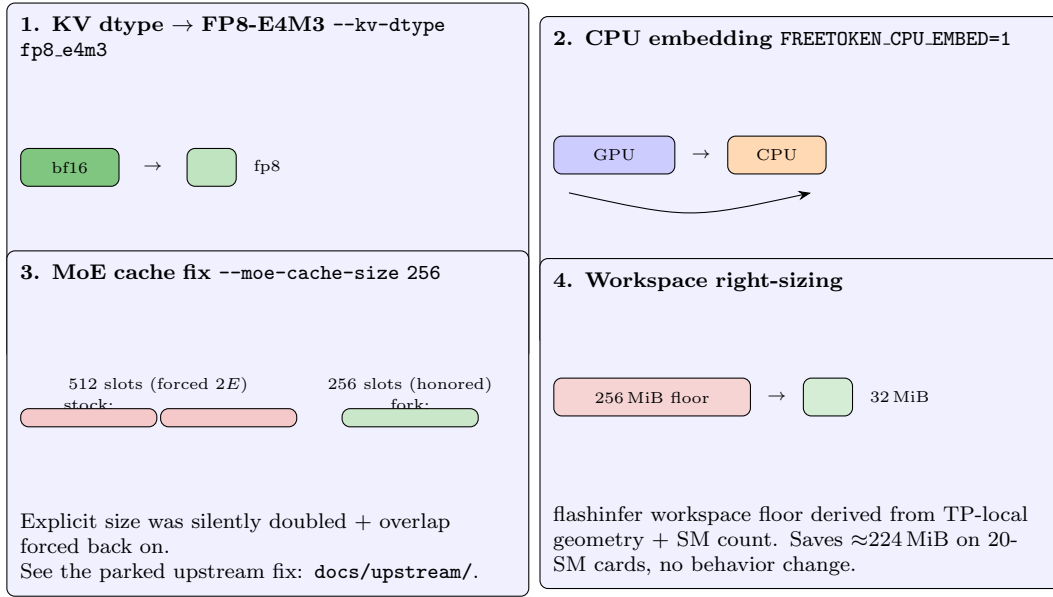
4 One decode step: the token’s journey



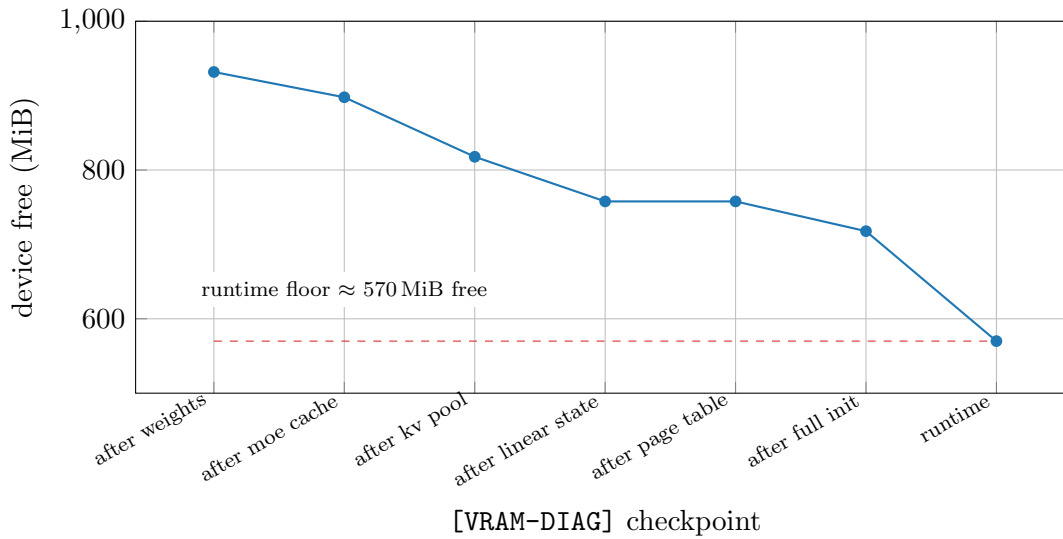
CUDA graphs OFF here: the CPU hop inside graph capture is illegal (unpinned copy) $\Rightarrow$ `--cuda-graph-max-bs 0`

The CPU embedding lookup and the CPU expert compute each cross PCIe once per token — cheap next to reading 8 expert weight sets. Measured decode: **10–14 tok/s** even at 65K context.

5 The four changes, at a glance



6 Memory ledger: every init checkpoint



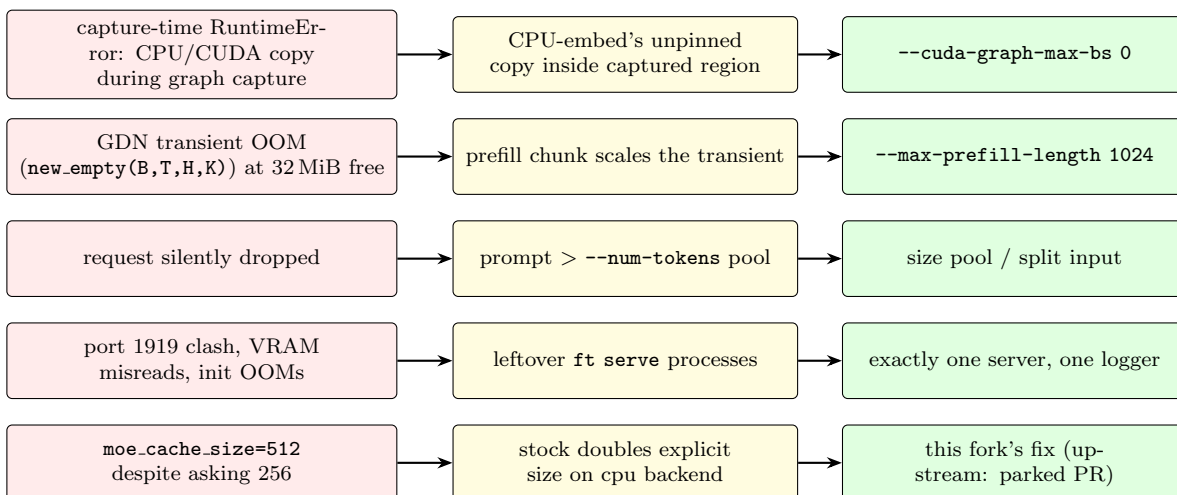
Big steps: weights (-3165 incl. embed before offload), KV pool -635 (FP8!), GDN linear state -60 , init tail -40 . This chart is exactly what `_vram.diag` prints — the diagnostic that found the 970 MiB embedding and the $256 \rightarrow 512$ cache doubling.

7 Measured performance

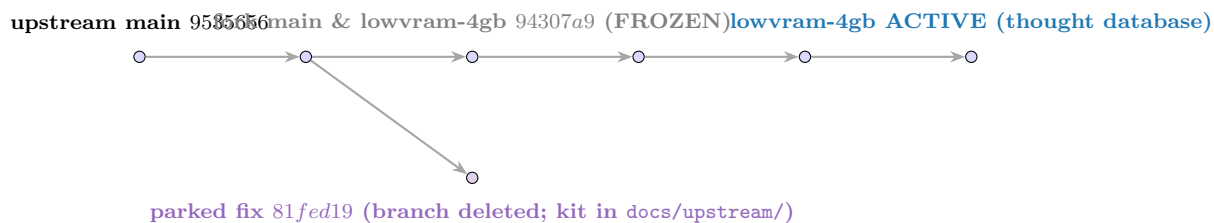


Prefill 206–230 tok/s (first chunk per request is warm-up at 5–50); decode 10–14 tok/s dominated by expert reads over PCIe. Soak test: one request grew to **65,349 tokens** of combined context — pool usage 1.00, no OOM, no quality collapse.

8 When it breaks: symptom → cause → fix



9 Branches and parked work (git topology)



Legend: the fork's `main` froze at 94307a9 (upstream + the 8-file 4-GiB patch + docs). All new work — including this thought database — lands on `lowvram-4gb`. The upstream bug-fix branch was parked: the full diff, test, and revival steps live in `docs/upstream/PARKED-RESOLVER-DESIGN.md`.

10 Regenerating this document

Action	Command
Rebuild the PDF	<code>make -C docs/visual</code>
Direct latexmk	<code>latexmk -pdf -interaction=nonstopmode -halt-on-error</code>
Toolchain	pdflatex + latexmk + tikz/pgfplots (Debian <code>texlive-pictures</code>)
Source of truth	this <code>.tex</code> ; the committed PDF is a convenience
LLM note	the TikZ sources are plain text — fully machine-readable